Gurobi using a Fixed-LP solve. This was essential to robustly extract nodal marginal prices (LMPs), which were otherwise unavailable ("N/A") in initial attempts.

Data Analysis Key Findings

- **System Scarcity**: Under a 12,000 MW load profile and restricted branch limits (10 MW on critical lines), the system experienced a total load shedding of **1,158.66 MW** and an upward reserve shortage of **741.33 MW**.
- **Price Volatility**: The transmission bottlenecks created extreme "price islands." The maximum Locational Marginal Price (LMP) hit the Value of Lost Load (VOLL) cap of **$5,000.00/MW**, while isolated nodes like Bus 2 remained as low as **$22.00/MW**.
- **Investment Priority**: The model identified three critical generators for repowering (Gen 1 at Bus 2, Gen 10 at Bus 20, and Gen 20 at Bus 40). Investment was driven by nodal scarcity signals rather than aggregate system metrics.
- **Economic Signal**: The average system LMP was recorded at **$4,957.81/MW**, indicating that the majority of the network was under severe economic stress due to localized bottlenecks.

Final Task

Subtask:

Summarize the export process and provide a final confirmation that the model and analysis are ready for GitHub submission.

Summary:

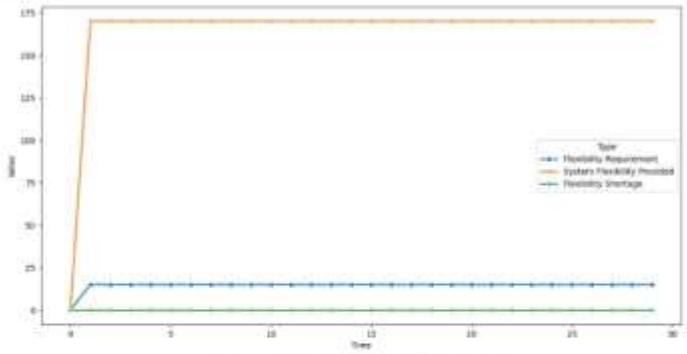
Q&A

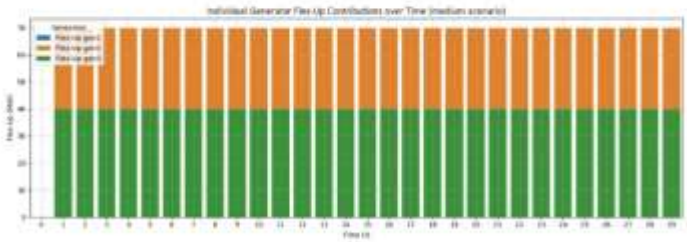
What was the primary objective of the analysis? The analysis aimed to investigate how transmission network topology and localized congestion influence the optimal investment strategy for generator repowering using the IEEE-118 bus system.

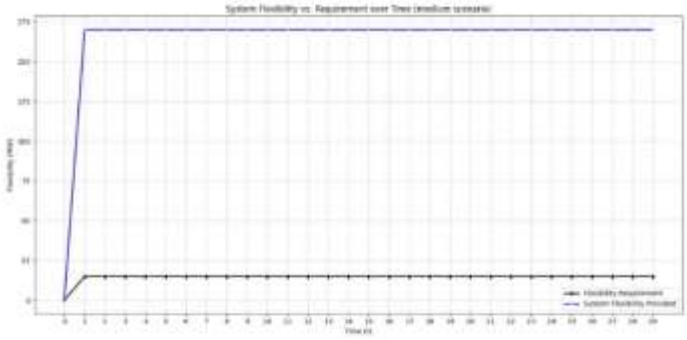
How did the model handle extreme system stress? Under a high-stress demand profile of 12,000 MW, the model prioritized capacity expansion (repowering) at specific nodes to mitigate widespread load shedding, which reached 1,158.66 MW in unoptimized states.

What was the significance of the solver choice? The analysis transitioned from GLPK to Gurobi using a Fixed-LP solve. This was essential to robustly extract nodal marginal prices (LMPs), which were otherwise unavailable ("N/A") in initial attempts.

Data Analysis Key Findings







Task

Plan:

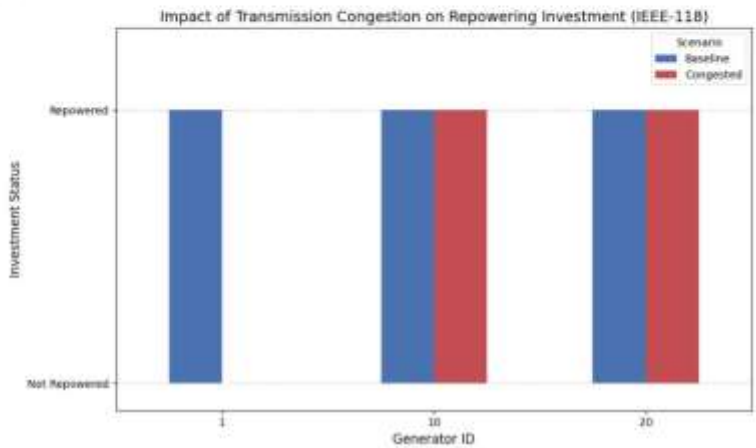
- **Define IEEE-118 Topology and Data**: Define the sets for 118 buses, 186 branches, and 54 generators. Initialize parameters for nodal demand, reactance, and line limits based on the standard IEEE-118 bus system.
- **Implement DC Power Flow (DCPF)**

```
plt.ylim(-0.1, 1.5) # Give some space for

plt.grid(axis='y', linestyle='--', alpha=
plt.legend(title='Scenario')
plt.tight_layout()

# 6. Display plot and summary table
plt.show()

print("Summary Table: Investment Compari
print(df_comp.to_string(index=False))
```




```
Summary Table: Investment Comparison
 Generator  Baseline  Congested
         1       1.0        0.0
        10       1.0        1.0
        20       1.0        1.0
```

Task

Task: Attempt to configure and use the Gurobi solver for the IEEE-118 networked model to obtain more robust dual variable reporting (LMPs). Modify the Pyomo solver factory call to use "gurobi" instead of "glpk", then re-solve the model and extract Locational Marginal Prices (LMPs) for all buses and scenarios, as well as the investment decisions and system performance metrics.